Analysis of Algorithms
Lecture 1

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Lectures Outline

- Introduction to Course
- Role of Algorithms
- Algorithm and its Attributes
- Analyzing algorithms
- Approaches to Analysis: Empirical Approach, Analytical
- Algorithm Specification
- RAM Computational Model
- Computational Complexity

Introduction to Course

- This course is an advanced undergraduate course focusing on the design and analysis of algorithms. It covers fundamental concepts such as the role of algorithms in computing, asymptotic analysis (Big-O, Big-Ω, Big-Θ), recursion and recurrence relations, and loop invariants, study of sorting and searching algorithms, string matching algorithms, heaps and hashing, graph traversal algorithms, and major algorithm design techniques including divide and conquer, greedy algorithms, and dynamic programming.

Introduction to Course

- Course Code:
  - CS: CSAA-341, AI: CSAA-226
- Credit Hours: [3+0]
- Prerequisite:
  - CS: [CSDS-208], AI: [CSDS-224]
- Text books:

  Introduction to Algorithms (3rd edition) by Thomas H. Corman, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein

  Algorithm Design (1st edition, 2013/2014) Jon Kleinberg, Eva Tardos

Why Study Algorithms?

- Multiple solutions exist for the same problem
- All correct algorithms are not equally efficient
- Efficiency matters for large inputs
- Real-world systems depend on fast algorithms

Role of Algorithms in Computer Science

- Core of problem solving
- Used in:
  - Data processing
  - Networking
  - Artificial Intelligence
  - Operating Systems
- Determines performance and feasibility

What is an Algorithm?

An algorithm is a **well-defined** and **effective** sequence of computation steps that takes some value, or set of values, as input and produces some value, or set of values, as output.

Example: sorting

- Input: A sequence of n numbers <a1,a2,a3…an>
- Output: A permutation (re-ordering) <b1,b2,b3…bn> of the input sequence such that b1<b2<b3…<bn

Attributes of an Algorithm

- Input
- Output
- Definiteness
- Finiteness
- Effectiveness
- Correctness

Problems solved by algorithms

- Sorting/searching are by no mean the only computational problem for which algorithms have been developed.
- Otherwise, we wouldn't have the whole course on this topic
- Practical application of algorithms are ubiquitous and include the following examples

Practical applications

- Internet world
- Electronic commerce
- Manufacturing and other commercial settings
- Shortest path
- Matrices multiplication order
- DNA sequence matching

Common about algorithms

- There are many candidate solutions, most of which are not what we want, finding one that we do want can present quite a challenge.
- There are practical applications (its not just mathematical exercises to develop algorithms.)

Why Analyze Algorithms?

- Predict performance before implementation
- Compare multiple algorithms
- Algorithms help us to understand scalability.
- Optimize resource usage
- Performance often draws the line between what is feasible and what is impossible.
- Algorithmic mathematics provides a language for talking about program behavior.
- Performance is the currency of computing.
- Speed is fun!

What Does Algorithm Analysis Measure?

- Time complexity
- Space complexity
- Best case
- Average case
- Worst case

Approaches to Algorithm Analysis

- Empirical Approach
- Analytical Approach

Empirical Approach

- Implement algorithm
- Run on test inputs
- Measure execution time

Limitations of Empirical Approach

- Machine dependent
- Input dependent
- Not scalable for large inputs
- Hard to generalize results

Analytical Approach

- Uses mathematical modeling
- Independent of machine
- Measures growth rate
- Predicts performance for large inputs

Algorithm Specification

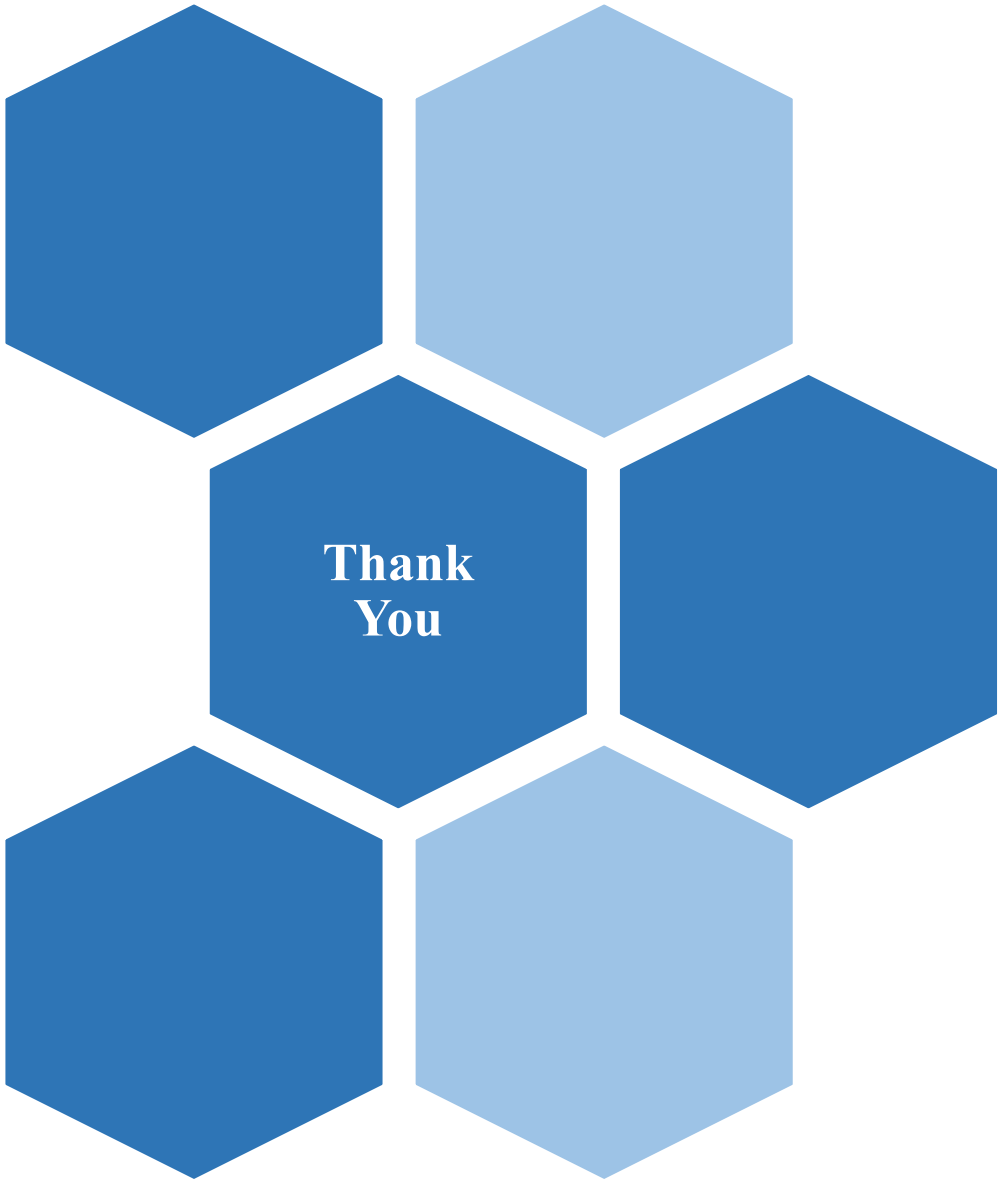
- Description of algorithm logic
- Written using:
  - Pseudocode
  - Flowcharts
- Independent of programming language

RAM Computational Model

- Random Access Machine (RAM)
- Assumes:
  - Constant time for basic operations
  - Sequential execution
- Simplifies analysis

Computational Complexity

- Measures algorithm efficiency
- Expressed as a function of input size
- Focus on growth rate
- Independent of hardware


Thank
You